

CSE-251

ELECTRONIC CIRCUITS

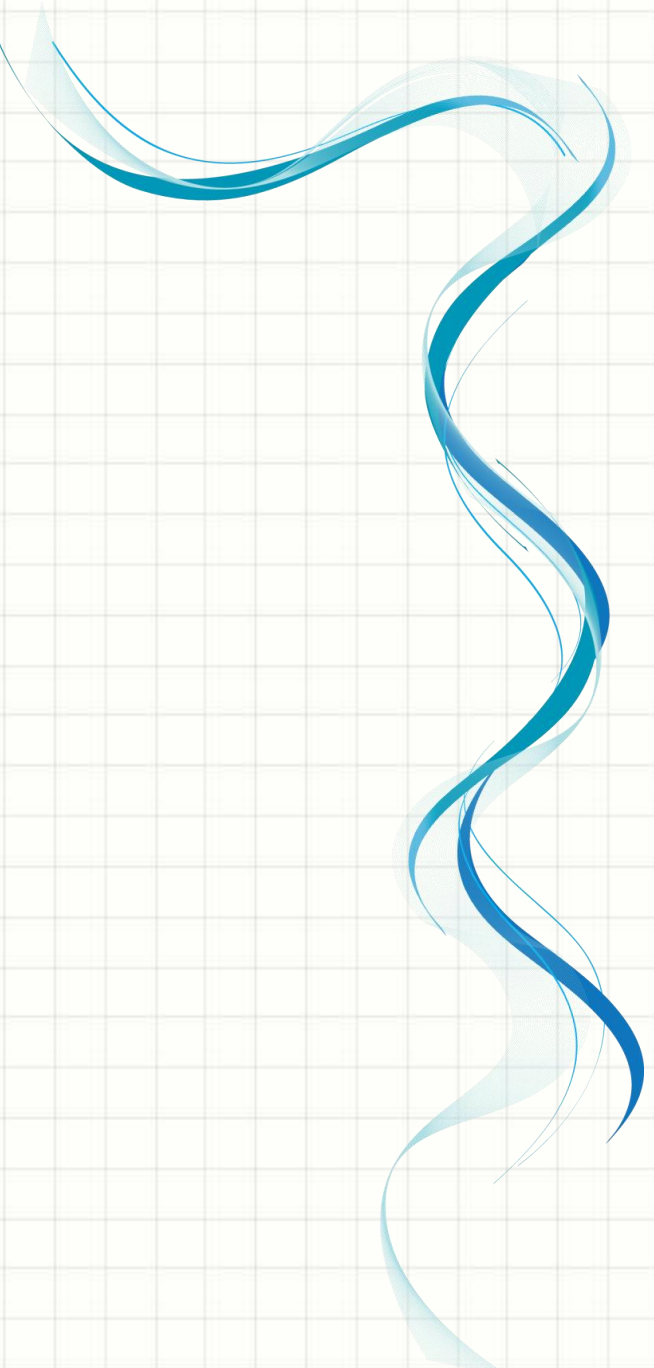
LECTURE-14

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Today's Overview:

- ❑ *Operational Amplifiers*
- ❑ *Inverting and Non-inverting Amplifiers*



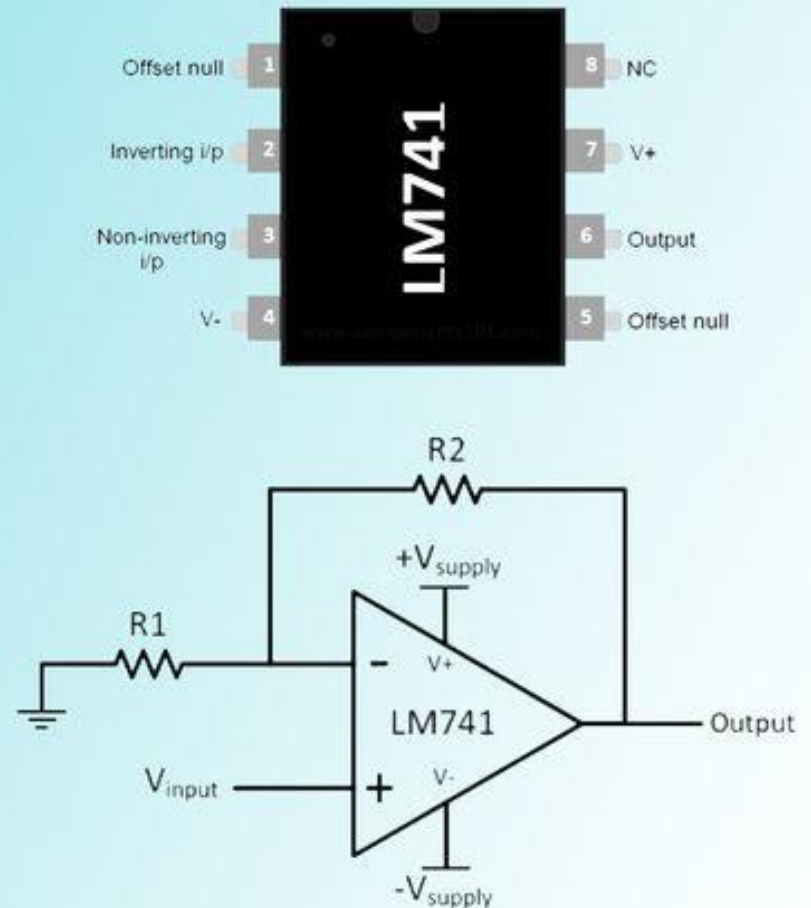
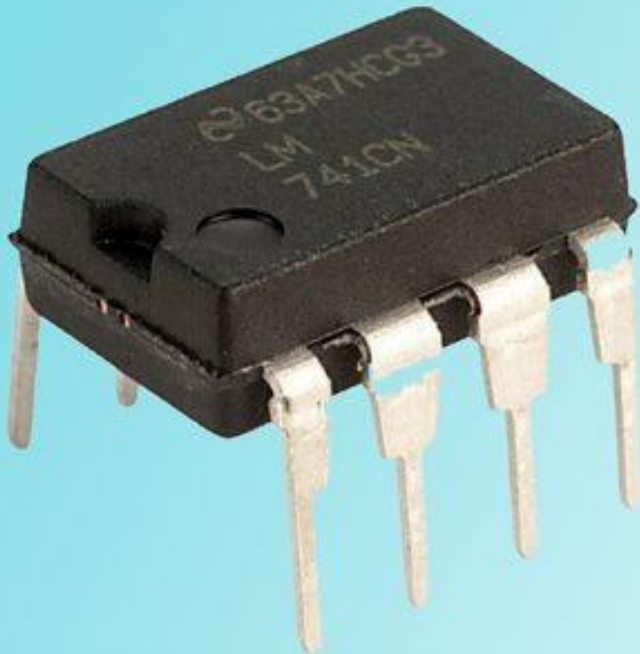
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Operational Amplifiers

OP-Amp:

An operational amplifier (OP-Amp) is a circuit that can perform such mathematical operations as addition, subtraction, integration and differentiation.

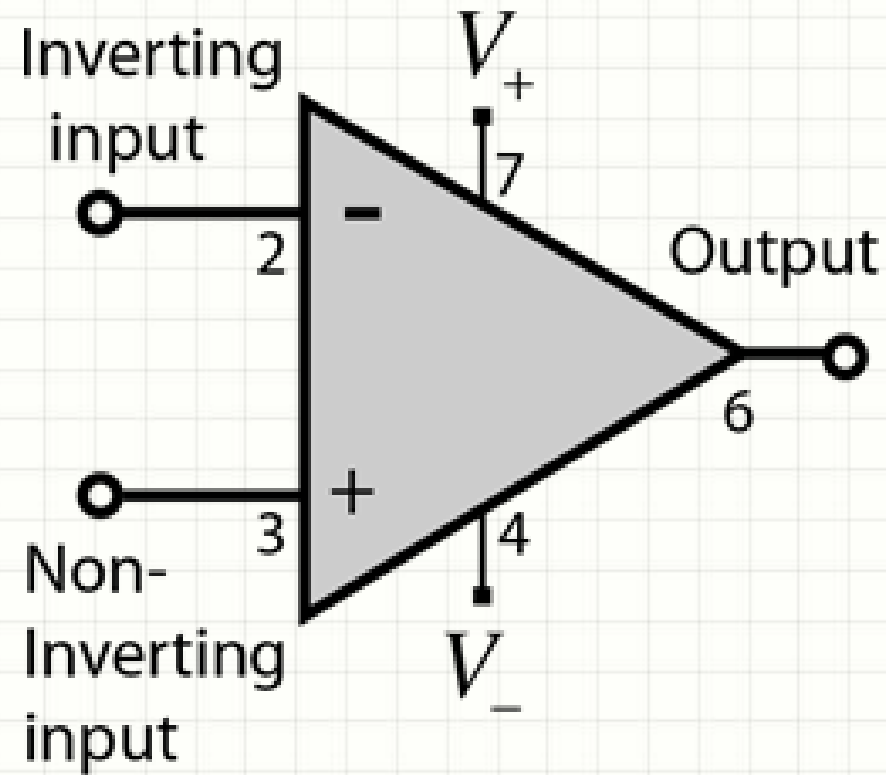
Op-Amp IC LM741



Applications:

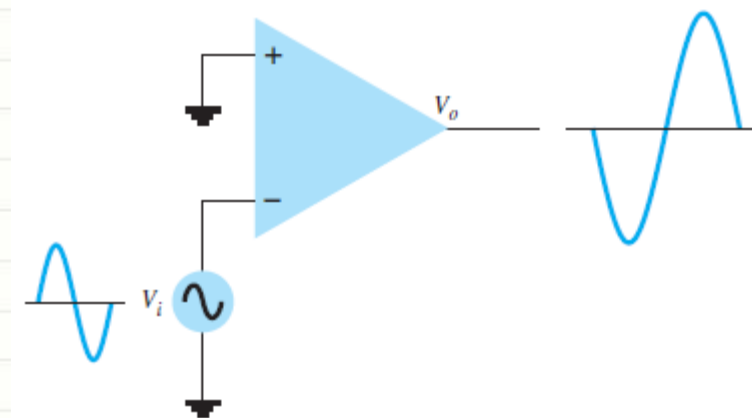
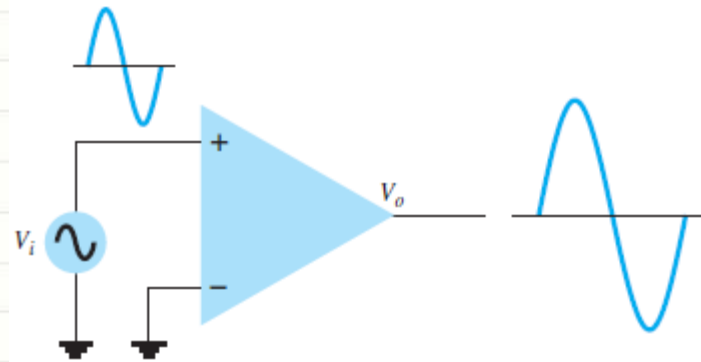
- Simple Amplifiers
- Summers
- Comparators
- Integrators
- Differentiators
- Analog to Digital Converters

Pin Output:



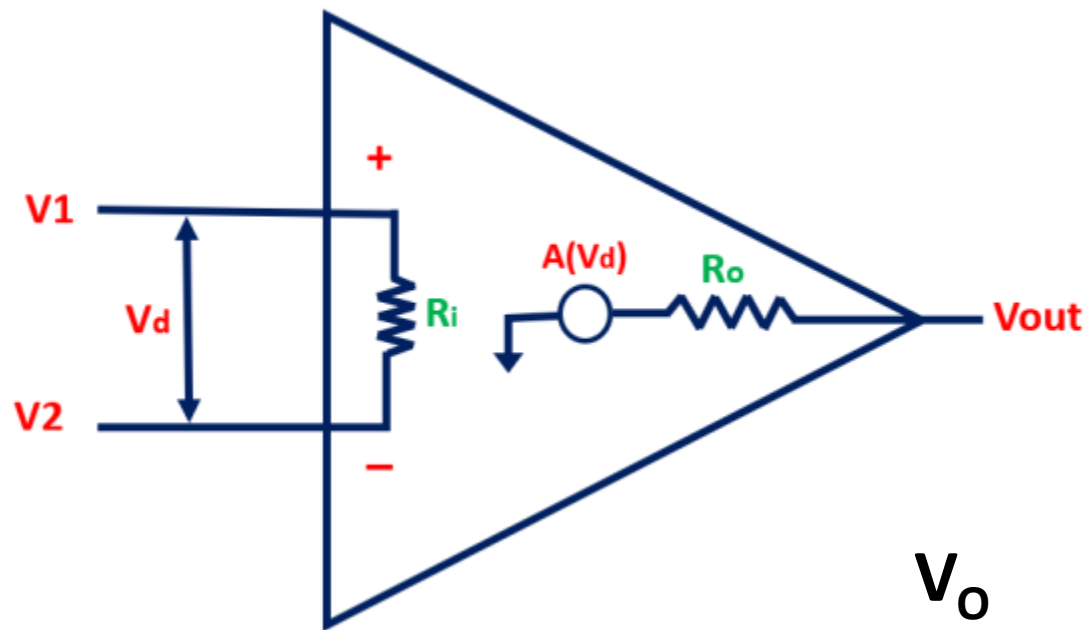
Differential Amplifier:

*A **differential amplifier** is a circuit that can accept two input signals and amplify the difference between these two input signals.*



Characteristics of an op-amp:

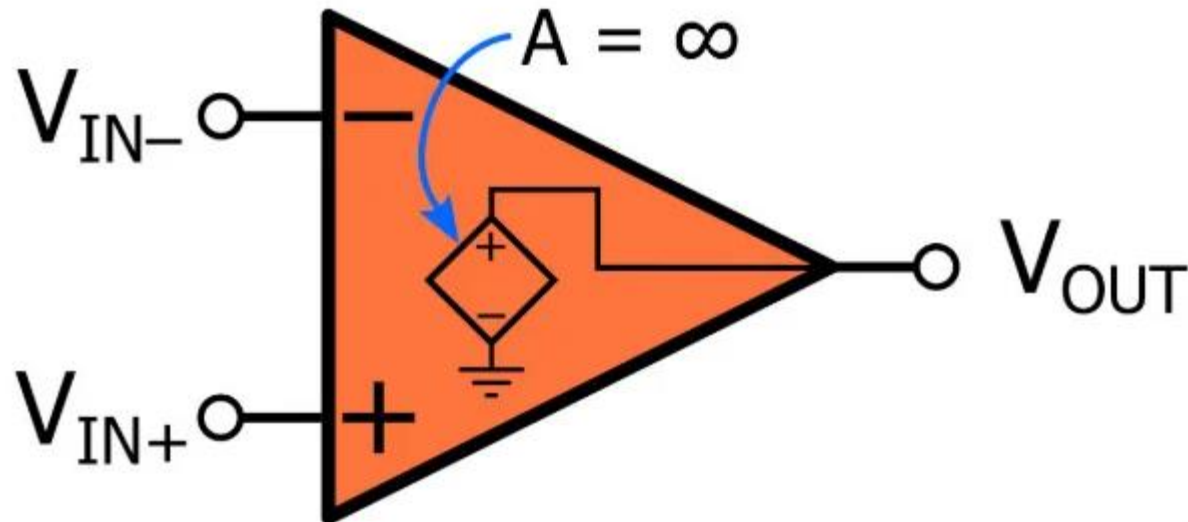
- ❑ High Input Impedance
- ❑ Low Output Impedance



$$\begin{aligned} V_o &= AV_d \\ &= A(V_1 - V_2) \end{aligned}$$

Characteristics of an op-amp:

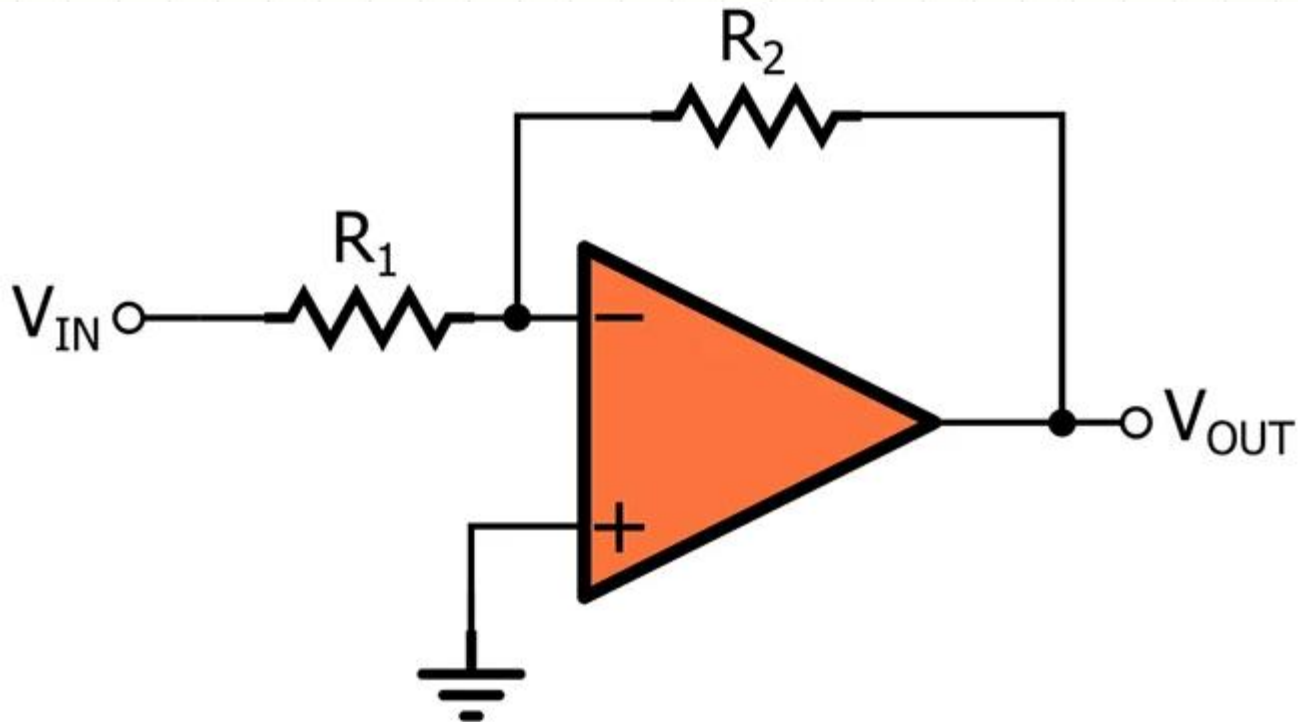
- ❑ Input resistance is infinity
- ❑ Output resistance is zero
- ❑ Voltage gain is infinity
- ❑ Band width is infinity



Inverting and Non-inverting Amplifiers



Inverting Amplifier:



$$\frac{V_o}{V_i} = -\frac{R_f}{R_i}$$

Problem:

EXAMPLE 10.5 If the circuit of Fig. 10.34 has $R_1 = 100 \text{ k}\Omega$ and $R_f = 500 \text{ k}\Omega$, what output voltage results for an input of $V_1 = 2 \text{ V}$?

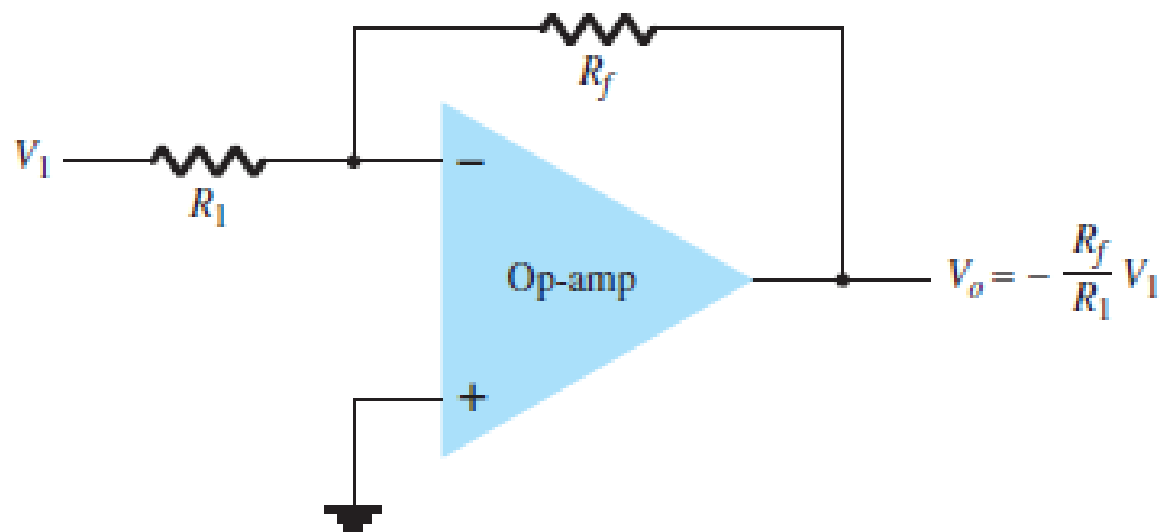


FIG. 10.34

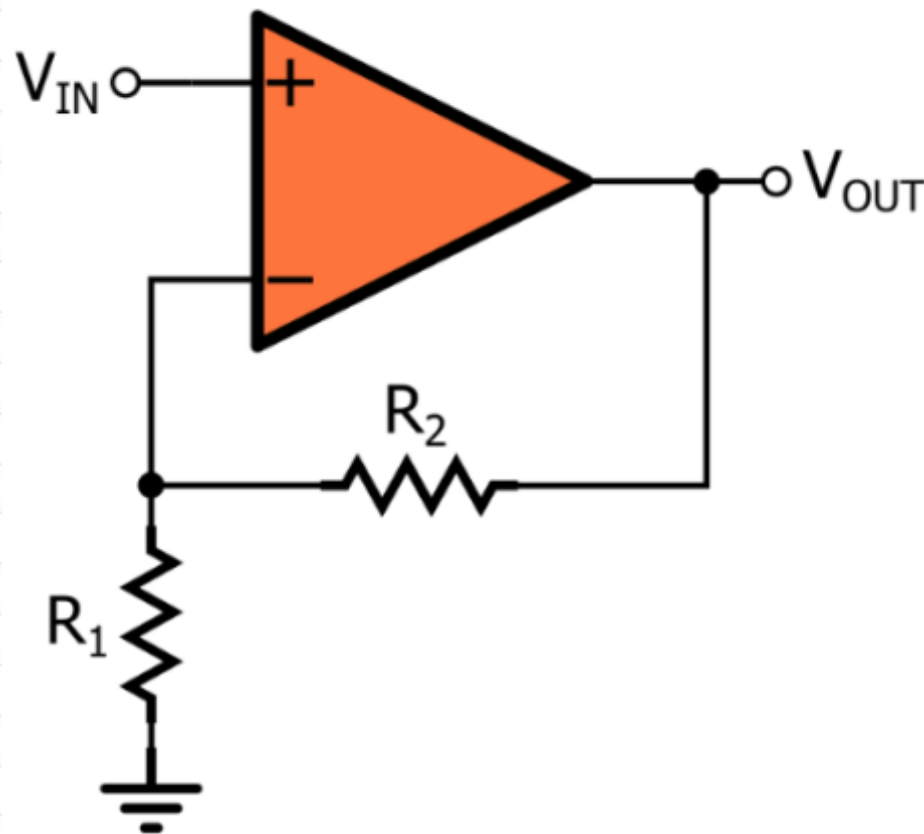
Inverting constant-gain multiplier.

Problem:

Solution:

$$\text{Eq. (10.8): } V_o = -\frac{R_f}{R_1} V_1 = -\frac{500 \text{ k}\Omega}{100 \text{ k}\Omega} (2 \text{ V}) = -10 \text{ V}$$

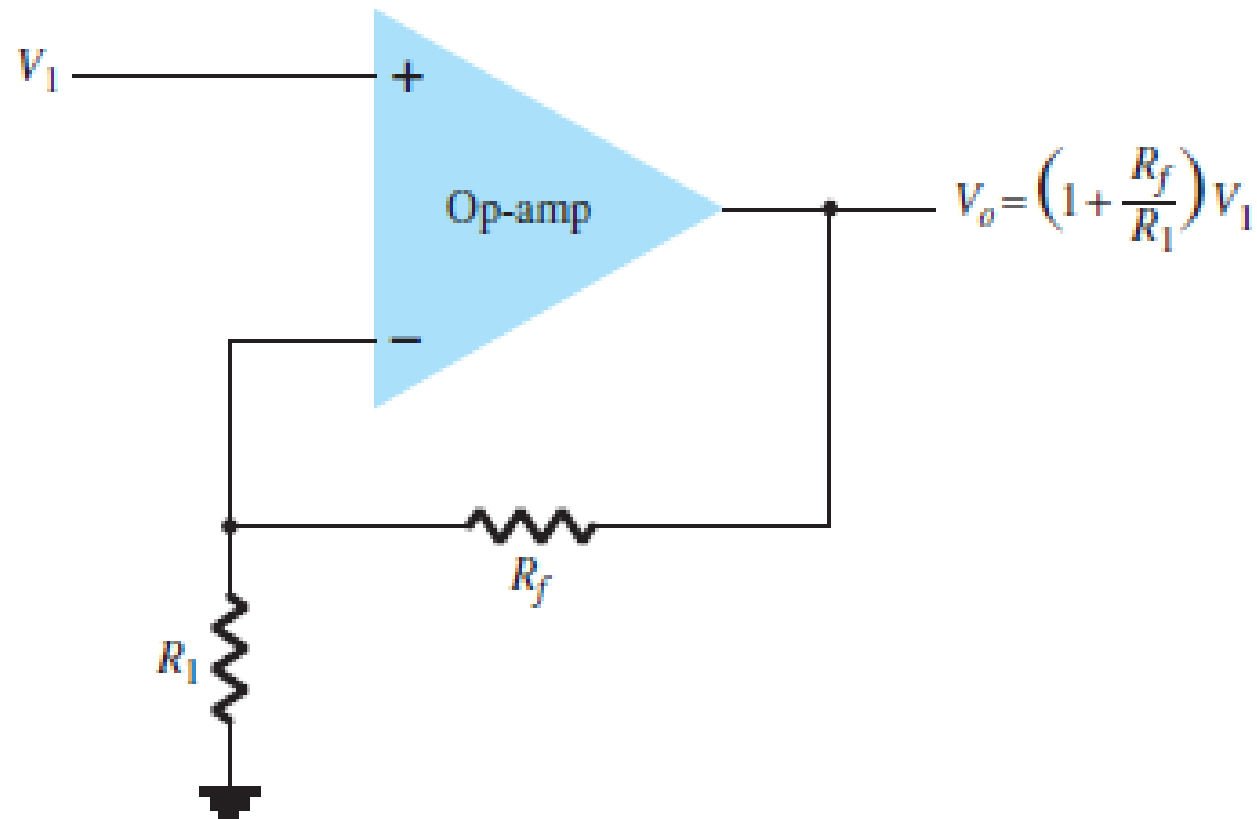
Non-inverting Amplifier:



$$\frac{V_o}{V_i} = \frac{R_1 + R_f}{R_1} = 1 + \frac{R_f}{R_1}$$

Problem:

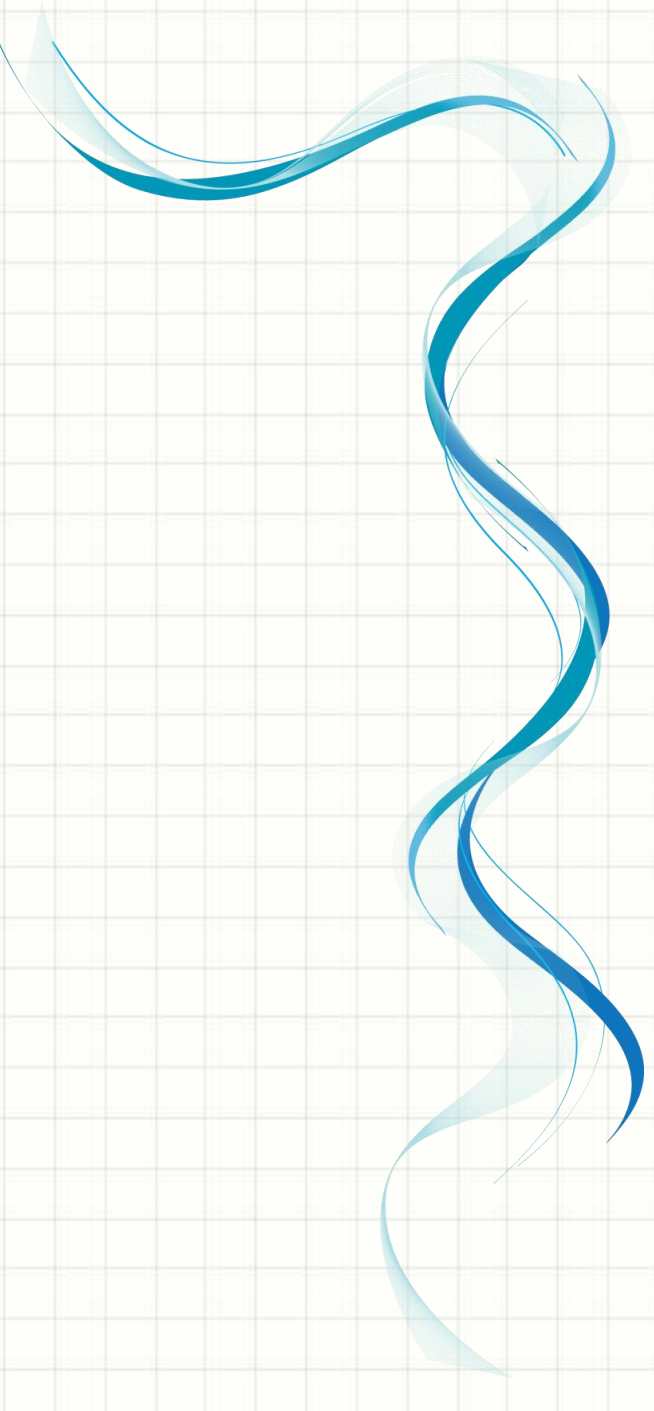
EXAMPLE 10.6 Calculate the output voltage of a noninverting amplifier (as in Fig. 10.35) for values of $V_1 = 2\text{ V}$, $R_f = 500\text{ k}\Omega$, and $R_1 = 100\text{ k}\Omega$.



Problem:

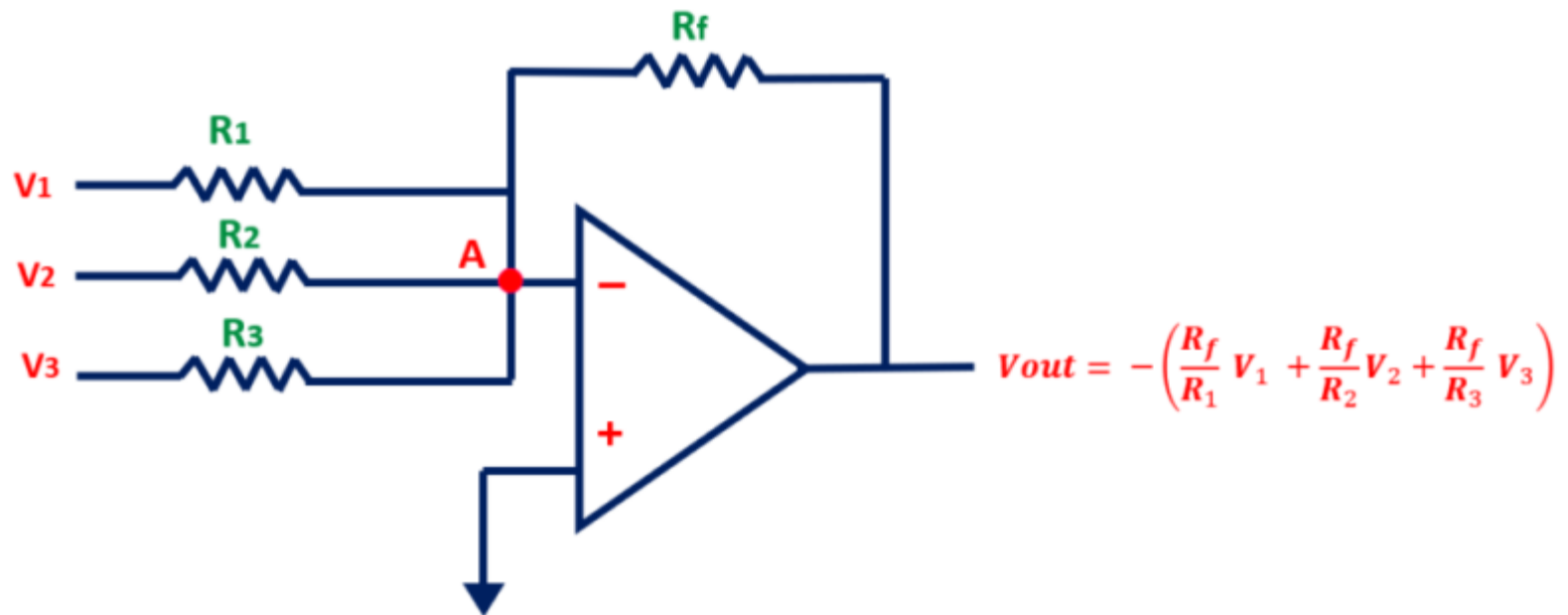
Solution:

$$\text{Eq. (10.9): } V_o = \left(1 + \frac{R_f}{R_1}\right)V_1 = \left(1 + \frac{500 \text{ k}\Omega}{100 \text{ k}\Omega}\right)(2 \text{ V}) = 6(2 \text{ V}) = +12 \text{ V}$$

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Summing Amplifier

Summing Amplifier/Unity Follower:



$$V_o = -\left(\frac{R_f}{R_1} V_1 + \frac{R_f}{R_2} V_2 + \frac{R_f}{R_3} V_3\right)$$

Problem:

EXAMPLE 10.7 Calculate the output voltage of an op-amp summing amplifier for the following sets of voltages and resistors. Use $R_f = 1 \text{ M}\Omega$ in all cases.

- a. $V_1 = +1 \text{ V}$, $V_2 = +2 \text{ V}$, $V_3 = +3 \text{ V}$, $R_1 = 500 \text{ k}\Omega$, $R_2 = 1 \text{ M}\Omega$, $R_3 = 1 \text{ M}\Omega$.
- b. $V_1 = -2 \text{ V}$, $V_2 = +3 \text{ V}$, $V_3 = +1 \text{ V}$, $R_1 = 200 \text{ k}\Omega$, $R_2 = 500 \text{ k}\Omega$, $R_3 = 1 \text{ M}\Omega$.

Problem:

Solution: Using Eq. (10.11), we obtain

$$\begin{aligned}\text{a. } V_o &= -\left[\frac{1000 \text{ k}\Omega}{500 \text{ k}\Omega} (+1 \text{ V}) + \frac{1000 \text{ k}\Omega}{1000 \text{ k}\Omega} (+2 \text{ V}) + \frac{1000 \text{ k}\Omega}{1000 \text{ k}\Omega} (+3 \text{ V}) \right] \\ &= -[2(1 \text{ V}) + 1(2 \text{ V}) + 1(3 \text{ V})] = -7 \text{ V}\end{aligned}$$

$$\begin{aligned}\text{b. } V_o &= -\left[\frac{1000 \text{ k}\Omega}{200 \text{ k}\Omega} (-2 \text{ V}) + \frac{1000 \text{ k}\Omega}{500 \text{ k}\Omega} (+3 \text{ V}) + \frac{1000 \text{ k}\Omega}{1000 \text{ k}\Omega} (+1 \text{ V}) \right] \\ &= -[5(-2 \text{ V}) + 2(3 \text{ V}) + 1(1 \text{ V})] = +3 \text{ V}\end{aligned}$$



QUESTIONS?

SEE YOU IN THE NEXT CLASS

